

Terminology

1. Exploratory Data Analysis (EDA)

Definition: EDA is the process of understanding a dataset using statistics and visualizations.

Why needed: It helps identify distributions, relationships, unusual values, and patterns before modeling.

When used: Use EDA before statistical testing and modeling.

Project observation: In this project, EDA was used to study age, BMI, children, charges, and their relationships.

```
insurance_df.describe()
insurance_df.info()
```

2. Mean and Median

Definition: Mean is the arithmetic average. Median is the middle value after sorting.

Why needed: Comparing them helps understand the center of the data and possible skewness.

When used: Use both when distributions may contain outliers.

Project observation: For charges, mean (\$13,270.42) is higher than median (\$9,382.03), supporting right skew.

```
insurance_df['charges'].mean()
insurance_df['charges'].median()
```

3. Standard Deviation

Definition: Standard deviation measures how spread out values are around the mean.

Why needed: A larger standard deviation means greater variability.

When used: Use it to describe numerical variables.

Project observation: Charges have a standard deviation of about \$12,110, showing substantial variation.

```
insurance_df['charges'].std()
```

4. IQR

Definition: Interquartile Range (IQR) is Q3 minus Q1.

Why needed: It describes the spread of the middle 50% and is less affected by extreme values.

When used: Use it when outliers or skewed data are present.

Project observation: The IQR of charges is about \$11,899.63.

```
q1 = insurance_df['charges'].quantile(0.25)
q3 = insurance_df['charges'].quantile(0.75)
iqr = q3 - q1
```

5. Skewness

Definition: Skewness describes the asymmetry of a distribution.

Why needed: Positive skew means a longer right tail; negative skew means a longer left tail.

When used: Use it to understand distribution shape and support test/model decisions.

Project observation: Charges have skewness 1.516, indicating strong positive skew.

```
insurance_df['charges'].skew()
```

6. Kurtosis

Definition: Kurtosis describes tail heaviness relative to a normal distribution. Pandas reports excess kurtosis.

Why needed: It helps identify distributions with unusually heavy or light tails.

When used: Use it when assessing distribution shape and residual normality.

Project observation: Charges have excess kurtosis 1.606, indicating heavier tails than a normal distribution.

```
insurance_df['charges'].kurtosis()
```

7. Histogram and KDE

Definition: A histogram shows frequencies across numeric intervals. KDE is a smooth estimate of the distribution.

Why needed: Together they make the distribution shape easier to see.

When used: Use for numerical distribution analysis.

Project observation: The charges distribution showed a clear long right tail.

```
sns.histplot(insurance_df['charges'], kde=True)  
plt.show()
```

8. Scatter Plot

Definition: A scatter plot displays the relationship between two numerical variables.

Why needed: It helps identify trends, spread, clusters, and outliers.

When used: Use for bivariate numerical analysis.

Project observation: Age vs charges showed a positive relationship, while BMI vs charges was weaker.

```
sns.scatterplot(data=insurance_df, x='age', y='charges')  
plt.show()
```

9. Correlation

Definition: Correlation measures the strength and direction of a linear relationship between numerical variables.

Why needed: It provides a quick numerical summary of linear association.

When used: Use when comparing numerical variables.

Project observation: Age had the strongest numerical correlation with charges (about 0.30). Correlation does not prove causation.

```
insurance_df[['age','bmi','children','charges']].corr()
```

10. Hypothesis Testing

Definition: Hypothesis testing uses sample data to evaluate a claim about a population.

Why needed: It provides a formal statistical decision using a test statistic and p-value.

When used: Use when comparing groups or testing relationships.

Project observation: Two hypothesis tests were performed in this project.

```
from scipy import stats
```

11. Null and Alternative Hypothesis

Definition: H0 represents the default/no-effect claim. H1 represents the competing claim.

Why needed: They define exactly what the statistical test is evaluating.

When used: State H0 and H1 before testing.

Project observation: For smokers vs non-smokers: H0 = no significant difference; H1 = significant difference.

```
alpha = 0.05
```

12. Significance Level (α)

Definition: Alpha is the threshold used to decide whether evidence is statistically significant.

Why needed: It controls the chosen significance criterion.

When used: A common value is 0.05; this lab uses 0.05.

Project observation: If $p < 0.05$, H0 is rejected; otherwise, H0 is not rejected.

```
alpha = 0.05
```

13. p-value

Definition: The p-value measures how unusual the observed result would be if H0 were true.

Why needed: It is used with alpha to make the statistical decision.

When used: Use it after calculating a hypothesis test.

Project observation: The smoker test had $p < 0.001$, so H_0 was rejected. A p-value is not the probability that H_0 is true.

```
statistic, p_value = stats.mannwhitneyu(smoker_charges, non_smoker_charges)
```

14. Shapiro-Wilk Test

Definition: Shapiro-Wilk tests whether data are consistent with a normal distribution.

Why needed: Normality can affect the choice of statistical test.

When used: Use it before choosing between parametric and non-parametric approaches when appropriate.

Project observation: Both smoker and non-smoker charge groups had $p < 0.05$, so normality was rejected.

```
stats.shapiro(smoker_charges)
stats.shapiro(non_smoker_charges)
```

15. Levene's Test

Definition: Levene's test checks whether group variances are equal.

Why needed: Equal variance is an assumption for some versions of parametric tests.

When used: Use it when comparing two or more groups.

Project observation: The smoker comparison had $p < 0.05$, indicating unequal variances.

```
stats.levene(smoker_charges, non_smoker_charges)
```

16. Mann-Whitney U Test

Definition: A non-parametric test for comparing two independent groups.

Why needed: It is useful when normality assumptions are not satisfied.

When used: Use for two independent groups when a non-parametric approach is appropriate.

Project observation: The test gave $U = 284133$ and $p < 0.001$; H_0 was rejected.

```
stats.mannwhitneyu(smoker_charges, non_smoker_charges, alternative='two-sided')
```

17. One-Way ANOVA

Definition: ANOVA tests whether the means of three or more groups are all equal.

Why needed: It provides one overall test instead of many separate pairwise tests.

When used: Use for a numerical outcome across 3+ categorical groups when ANOVA assumptions are appropriate.

Project observation: Region ANOVA gave $F = 2.9696$ and $p = 0.030893$, so H_0 was rejected.

```
stats.f_oneway(group1, group2, group3, group4)
```

18. OLS Regression

Definition: Ordinary Least Squares estimates a linear relationship between a numerical target and predictors.

Why needed: It quantifies how predictors are associated with the target while controlling for other predictors.

When used: Use when a linear regression model is appropriate.

Project observation: The target was charges and predictors included age, BMI, children, sex, smoker, and region.

```
X = sm.add_constant(X)
ols_model = sm.OLS(y, X).fit()
```

19. Dummy Variables

Definition: Dummy encoding converts categorical values into numeric indicator variables.

Why needed: OLS requires numeric predictors.

When used: Use when categorical predictors are included in regression.

Project observation: Reference categories were female, non-smoker, and northeast.

```
pd.get_dummies(regression_df, columns=['sex','smoker','region'], drop_first=True)
```

20. Regression Coefficient

Definition: A coefficient represents the expected change in the target for a one-unit predictor change, holding other predictors constant.

Why needed: Coefficients help interpret predictor effects.

When used: Use when interpreting regression models.

Project observation: Age coefficient ≈ 256.86 means charges increase by about \$256.86 per additional year, holding other variables constant.

```
ols_model.params
```

21. Confidence Interval

Definition: A confidence interval gives a range of plausible values for a population parameter under the model assumptions.

Why needed: It communicates uncertainty around an estimate.

When used: Use when interpreting regression coefficients or predictions.

Project observation: The 95% CI for age was approximately \$233.51 to \$280.20 per year.

```
ols_model.conf_int(alpha=0.05)
```

22. R² and Adjusted R²

Definition: R² is the proportion of target variation explained by the model. Adjusted R² accounts for model size.

Why needed: They summarize model explanatory performance.

When used: Use when evaluating linear regression.

Project observation: R² = 0.751 and adjusted R² = 0.749, so the model explains about 75.1% of variation.

```
ols_model.rsquared  
ols_model.rsquared_adj
```

23. Residuals and Fitted Values

Definition: A residual is actual value minus predicted value. A fitted value is a model prediction for an observed row.

Why needed: Residuals help evaluate model assumptions.

When used: Use residual plots to diagnose regression problems.

Project observation: Residuals were centered around zero but showed non-constant spread.

```
fitted_values = ols_model.fittedvalues  
residuals = ols_model.resid
```

24. Heteroscedasticity

Definition: Heteroscedasticity means residual variance changes across fitted values.

Why needed: It can affect standard errors and inference.

When used: Check using residual plots and appropriate statistical diagnostics.

Project observation: The residuals-vs-fitted plot showed some evidence of non-constant variance.

```
plt.scatter(fitted_values, residuals)  
plt.axhline(0, linestyle='--')
```

25. Q-Q Plot

Definition: A Q-Q plot compares observed residual quantiles with theoretical normal quantiles.

Why needed: It helps assess whether residuals are approximately normally distributed.

When used: Use as a regression diagnostic.

Project observation: The residual Q-Q plot deviated from the reference line, especially in the tails.

```
sm.qqplot(residuals, line='45', fit=True)  
plt.show()
```

26. Omnibus Test

Definition: The Omnibus test checks whether residuals depart from normality.

Why needed: It provides a formal normality diagnostic.

When used: Use for regression residual diagnostics.

Project observation: The model output showed Omnibus $p < 0.001$, indicating non-normal residuals.

```
from statsmodels.stats.stattools import omni_normtest  
omni_normtest(residuals)
```

27. Jarque-Bera Test

Definition: Jarque-Bera uses skewness and kurtosis to assess normality.

Why needed: It provides another formal residual-normality check.

When used: Use alongside Q-Q plots and other diagnostics.

Project observation: JB ≈ 718.887 with $p < 0.001$, supporting non-normality.

```
from statsmodels.stats.stattools import jarque_bera  
jarque_bera(residuals)
```

28. Durbin-Watson

Definition: Durbin-Watson checks residual autocorrelation, especially first-order correlation.

Why needed: It helps assess whether residuals are correlated with nearby observations.

When used: Use as a regression diagnostic, particularly for ordered data.

Project observation: DW ≈ 2.088 , indicating no strong autocorrelation.

```
from statsmodels.stats.stattools import durbin_watson  
durbin_watson(residuals)
```

29. VIF and Multicollinearity

Definition: VIF measures how much a predictor's variance is inflated by relationships with other predictors.

Why needed: High multicollinearity can make coefficient estimates unstable.

When used: Use VIF when multiple predictors are used in regression.

Project observation: All VIF values were below 2; the highest was about 1.65, indicating no serious multicollinearity.

```
from statsmodels.stats.outliers_influence import variance_inflation_factor
```

30. Prediction Interval

Definition: A prediction interval gives a range for a future individual observation.

Why needed: It accounts for uncertainty in the estimated mean and individual outcome variation.

When used: Use when predicting an individual case.

Project observation: The Streamlit app displays a 95% prediction interval for predicted insurance charges.

```
prediction = loaded_model.get_prediction(input_data)
prediction.summary_frame(alpha=0.05)
```

31. Joblib / Model Persistence

Definition: Model persistence means saving a fitted model so it can be reused later.

Why needed: It prevents retraining every time the application starts.

When used: Use when deploying trained models in applications.

Project observation: The fitted OLS model and feature list were saved as models/ols_model.pkl.

```
joblib.dump(model_data, 'models/ols_model.pkl')
loaded_model_data = joblib.load('models/ols_model.pkl')
```

32. Streamlit

Definition: Streamlit is a Python framework for building interactive data applications.

Why needed: It allows analysis and models to be presented through a web interface.

When used: Use for simple interactive dashboards and data apps.

Project observation: The app contains Data Exploration, Hypothesis Testing Lab, and Live Prediction & Diagnostics tabs.

```
python -m streamlit run app/app.py
```

33. Plotly

Definition: Plotly is a library for interactive charts.

Why needed: Interactive charts allow users to inspect data dynamically.

When used: Use when dashboards need interactive visualizations.

Project observation: The Streamlit app uses Plotly for histograms, scatter plots, box plots, and correlation visualization.

```
import plotly.express as px
```